Lecture 13

Radiation

A compact source sends energy through increasingly large spheres. The observation time must increase with the radius to follow that energy. We compute the limiting outward flux of the retarded Maxwell field.

Objectives. *We derive the energy balance directly from Maxwell's equations and prove the far-field power formula. We apply it to a slowly varying neutral dipole source and distinguish an absolute error from a relative one.*

By the end of this lecture you will be able to:

- Derive the Poynting balance with its source-work sign.
- Compute the radiated power from a smooth compact conserved current.
- Obtain the dipole coefficient and control the slow-time error.

13.1 Observation time and energy flux

Use the real fields and units of Lectures 9 and 10. Write $r = |x|$ and $n = x/r \in \mathbb{S}^2$ when $x \neq 0$. The sphere area measure $\mathrm{d}\Omega$ has total area 4π.

Definition 13.1 (Far zone and radiated power)**.** The *retarded time* of an observation is $u = t - r$. The *far-zone limit* is $r \to \infty$ at fixed u and n. For the outward energy flux $\mathbf{E} \times \mathbf{B}$, the *radiated power* is the limit, when it exists,

$$P(u) = \lim_{r\to\infty} r^2 \int_{\mathbb{S}^2} (\mathbf{E} \times \mathbf{B})(r + u, rn) \cdot n \, \mathrm{d}\Omega. \tag{13.1}$$

For completeness, recall the Maxwell stress tensor in the convention of *Topology and Geometry*, Lecture 19. Writing $F_{\mathrm{em}} = \frac{1}{2} F_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$, it is

$$T_{\mu\nu} = -F_{\mu\alpha} F_\nu{}^\alpha + \frac{1}{4} \eta_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

The coefficients $F_{0i} = -E_i$ and $F_{ij} = \epsilon_{ijk} B_k$ give its energy components by contraction:

$$\rho_{\mathrm{en}} = T^{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2), \qquad T^{i0} = (\mathbf{E} \times \mathbf{B})_i.$$

Here ϵ_{ijk} is the alternating symbol, with $\epsilon_{123} = 1$. We prove the required balance without importing a stress-tensor conservation theorem.

Maxwell's equations give $\mathbf{E}_t = \operatorname{curl} \mathbf{B} - \mathbf{J}$ and $\mathbf{B}_t = -\operatorname{curl} \mathbf{E}$. The component product rule gives $\operatorname{div}(\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot \operatorname{curl} \mathbf{E} - \mathbf{E} \cdot \operatorname{curl} \mathbf{B}$. Taking the two scalar products yields the *Poynting identity*

$$\partial_t \rho_{\mathrm{en}} = \mathbf{E} \cdot (\operatorname{curl} \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot \operatorname{curl} \mathbf{E} = -\operatorname{div}(\mathbf{E} \times \mathbf{B}) - \mathbf{J} \cdot \mathbf{E}. \tag{13.2}$$

Thus $\mathbf{J}\cdot\mathbf{E}$ is the rate of energy transfer from the field to the prescribed source. Outside its support, the source-work term in the Poynting identity vanishes.

13.2 The far-field limit

Assume ρ and $\mathbf{J}$ are smooth, compactly supported in spacetime, and satisfy $\rho_t + \operatorname{div}\mathbf{J} = 0$. Use the retarded potentials from Lecture 6:

$$\phi_{\mathrm{em}}(t,x) = \frac{1}{4\pi}\int \frac{\rho(t-|x-y|,y)}{|x-y|}\,\mathrm{d}^3y, \qquad \mathbf{A}(t,x) = \frac{1}{4\pi}\int \frac{\mathbf{J}(t-|x-y|,y)}{|x-y|}\,\mathrm{d}^3y.$$

Derivatives pass to the smooth source as proved there. Their Lorenz function is $\ell = -G_{\mathrm{ret}} * (\rho_t + \operatorname{div}\mathbf{J}) = 0$. The wave equations and the reduction in Lecture 10 therefore give Maxwell's equations.

Theorem 13.2 (Radiated power). *For this retarded field define the vector and scalar amplitudes by*

$$\mathcal{A}(u,n) = \frac{1}{4\pi}\int \mathbf{J}(u+n\cdot y,y)\,\mathrm{d}^3y, \qquad \mathcal{V}(u,n) = \frac{1}{4\pi}\int \rho(u+n\cdot y,y)\,\mathrm{d}^3y. \tag{13.3}$$

Primes denote derivatives in u. With $P_{\mathrm{T}}(n) = \mathbb{1} - nn^{\mathsf{T}}$,

$$\begin{aligned}
\mathbf{A}(r+u,rn) &= \frac{\mathcal{A}(u,n)}{r} + O(r^{-2}), \qquad \phi_{\mathrm{em}}(r+u,rn) = \frac{\mathcal{V}(u,n)}{r} + O(r^{-2}),\\
\mathbf{E}(r+u,rn) &= -\frac{P_{\mathrm{T}}(n)\mathcal{A}'(u,n)}{r} + O(r^{-2}),\\
\mathbf{B}(r+u,rn) &= -\frac{n\times\mathcal{A}'(u,n)}{r} + O(r^{-2}).
\end{aligned} \tag{13.4}$$

The errors are uniform in $u\in\mathbb{R}$ and $n\in\mathbb{S}^2$. The limit (13.1) exists and equals

$$P(u) = \int_{\mathbb{S}^2} |n\times\mathcal{A}'(u,n)|^2\,\mathrm{d}\Omega. \tag{13.5}$$

Proof. We estimate the potentials and their derivatives before computing the flux. Enclose the spatial source support in $B_R(0)$ and take $r > 2R$. For $|y|\le R$, put $d = |rn-y|$ and $e = (rn-y)/d$. Taylor expansion of $\sqrt{1-2n\cdot y/r + |y|^2/r^2}$ on this bounded set gives

$$d = r - n\cdot y + O(r^{-1}), \qquad d^{-1} = r^{-1} + O(r^{-2}), \qquad e = n + O(r^{-1}). \tag{13.6}$$

These bounds are uniform in n and y. Figure 13.1 shows why the retarded source time approaches $u + n\cdot y$.

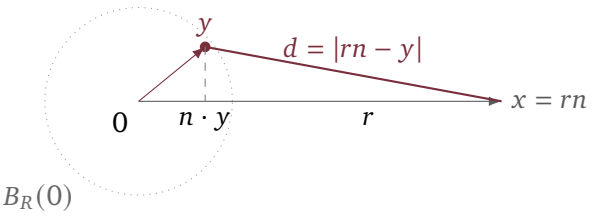


Figure 13.1: The source point y and observer rn have separation d. The projection $n\cdot y$ shortens the leading travel distance, so the source is sampled at time $u + n\cdot y$ in the limit.

Let V_f be the retarded potential of any one source component f, and let $\mathcal{V}_f(u,n) = (4\pi)^{-1}\int f(u+n\cdot y,y)\mathrm{d}^3y$. All derivatives of f are bounded, and the integration region is fixed. The mean-value estimate and (13.6) first give $V_f(r+u,rn) = \mathcal{V}_f(u,n)/r + O(r^{-2})$. Differentiating the exact integral at fixed (t,x) gives

$$\begin{aligned}\partial_t V_f(t,rn) &= \frac{1}{4\pi}\int \frac{f_t(t-d,y)}{d}\,\mathrm{d}^3y = \frac{\mathcal{V}_f'(u,n)}{r} + O(r^{-2}),\\ \partial_{x_i} V_f(t,rn) &= -\frac{1}{4\pi}\int e_i\left(\frac{f_t(t-d,y)}{d} + \frac{f(t-d,y)}{d^2}\right)\mathrm{d}^3y = -\frac{n_i\mathcal{V}_f'(u,n)}{r} + O(r^{-2}),\end{aligned} \tag{13.7}$$

where the asymptotic equalities are evaluated at $t = r+u$. Bounded second time derivatives justify the same mean-value estimates here. In particular, the radial derivative at fixed t has leading term $-\mathcal{V}_f'/r$. For a unit tangent vector v to the direction sphere, $D_n d[v] = -r(v\cdot y)/d = O(1)$. Differentiating the integral in this direction gives $D_n V_f = O(r^{-1})$. Its contribution to the Cartesian gradient has an additional factor $1/r$ and is therefore $O(r^{-2})$. Thus no angular derivative of the leading amplitude has been discarded at order $1/r$.

Current conservation relates the amplitudes. The total y-divergence of the compactly supported vector $\mathbf{J}(u+n\cdot y,y)$ integrates to zero. The chain rule gives

$$0 = \int \left[\operatorname{div}_y \mathbf{J} + n\cdot\partial_t\mathbf{J}\right](u+n\cdot y,y)\,\mathrm{d}^3y.$$

Combining this equality with $\rho_t = -\operatorname{div}_y\mathbf{J}$ proves $\mathcal{V}' = n\cdot\mathcal{A}'$. Substitution into (13.7) gives $\mathbf{E} = (n\mathcal{V}' - \mathcal{A}')/r + O(r^{-2})$ and $\mathbf{B} = -n\times\mathcal{A}'/r + O(r^{-2})$. These are (13.4).

Finally, set $b = \mathcal{A}'(u,n)$ and $E_* = -P_{\mathrm{T}}(n)b$. The magnetic leading coefficient is $n\times E_* = -n\times b$. Since $E_*\cdot n = 0$, the vector triple-product identity gives

$$r^2(\mathbf{E}\times\mathbf{B})(r+u,rn)\cdot n = \left[E_*\times(n\times E_*)\right]\cdot n + O(r^{-1}) = |n\times b|^2 + O(r^{-1}).$$

The error is uniform on the sphere. Its integral tends to zero, proving (13.5). ∎

13.3 A slowly varying neutral source

For a compact charge distribution of zero total charge, the *electric dipole moment* is $\mathbf{p}(t) = \int y\rho(t,y)\mathrm{d}^3y$. Choose fixed real $f\in C_c^\infty(\mathbb{R}^3)$ with $\int f\mathrm{d}^3y = 1$ and fixed $\mathbf{p}_0\in C_c^\infty(\mathbb{R};\mathbb{R}^3)$. For $0<\varepsilon\le 1$, consider

$$\mathbf{p}(t) = \mathbf{p}_0(\varepsilon t), \qquad \rho(t,y) = -\mathbf{p}(t)\cdot\nabla f(y), \qquad \mathbf{J}(t,y) = \dot{\mathbf{p}}(t)f(y). \tag{13.8}$$

This source is smooth and compactly supported in spacetime for each ε. Direct differentiation gives $\rho_t + \operatorname{div}\mathbf{J} = 0$. Coordinate integration by parts gives $\int\rho = 0$ and $\int y_i\rho\mathrm{d}^3y = p_i(t)\int f\mathrm{d}^3y = p_i(t)$, so the prescribed vector is indeed its dipole moment.

Example 13.3 (Dipole power with an absolute error)**.** For the source (13.8), compute the leading radiated power as $\varepsilon\downarrow 0$. Its amplitude derivative is

$$\mathcal{A}'(u,n) = \frac{1}{4\pi}\int \ddot{\mathbf{p}}(u+n\cdot y)f(y)\,\mathrm{d}^3y.$$

Since $\|\ddot{\mathbf{p}}\|_\infty = O(\varepsilon^2)$ and $\|\mathbf{p}^{(3)}\|_\infty = O(\varepsilon^3)$, the mean-value estimate gives

$$\mathcal{A}'(u,n) = \frac{\ddot{\mathbf{p}}(u)}{4\pi} + R_\varepsilon(u,n), \qquad |R_\varepsilon| \le \frac{\varepsilon^3\|\mathbf{p}_0^{(3)}\|_\infty}{4\pi}\int |y|\,|f(y)|\,\mathrm{d}^3y.$$

This estimate is uniform in u and n. The cross term in the squared amplitude is $O(\varepsilon^5)$, and its remainder square is $O(\varepsilon^6)$. To integrate the leading term, rotate a fixed vector $\mathbf{a}$ to the polar axis:

$$\int_{\mathbb{S}^2} |n \times \mathbf{a}|^2 \, \mathrm{d}\Omega = 2\pi |\mathbf{a}|^2 \int_{-1}^{1} (1 - s^2) \, \mathrm{d}s = \frac{8\pi}{3} |\mathbf{a}|^2.$$

The power formula therefore yields the rationalised *Larmor formula* with its slow-time error:

$$P(u) = \frac{|\ddot{\mathbf{p}}(u)|^2}{6\pi} + O(\varepsilon^5) \qquad \text{uniformly in } u \in \mathbb{R}. \tag{13.9}$$

The implied constant depends on the fixed profiles f and $\mathbf{p}_0$. The estimate is absolute. Dividing it by $|\ddot{\mathbf{p}}(u)|^2$ is invalid at a zero of the acceleration.

13.4 The limits of the source model

Retardation is a condition on the solution, as in Lecture 6. Adding a homogeneous Maxwell field preserves the source equations and can change the flux through infinity. Formula (13.5) describes the retarded field of the stated smooth source. The planned QFT course returns to choices of propagator and boundary conditions.

A point charge is outside these smooth-source hypotheses. The Abraham–Lorentz–Dirac self-force equation has runaway solutions. We do not derive or use it here. See Spohn, *Dynamics of Charged Particles and their Radiation Field* (2004), Chapter 9, and his study of the Lorentz–Dirac equation. The prescribed smooth current in this lecture is not a point particle acted on by its own singular field.

Summary. The Poynting vector is $\mathbf{E} \times \mathbf{B}$ and its balance includes $-\mathbf{J} \cdot \mathbf{E}$. A smooth compact conserved current has retarded far-field power $\int |n \times \mathcal{A}'|^2 \mathrm{d}\Omega$. Conservation removes the longitudinal electric coefficient. For the neutral slow-time dipole source, the leading power is $|\ddot{\mathbf{p}}|^2/(6\pi)$ with a uniform absolute $O(\varepsilon^5)$ error. This formula does not describe a singular self-force problem.

Self-check. At a fixed direction, suppose $\mathcal{A}'(u, n)$ is parallel to n. What are the leading radiation fields and flux in that direction?

Answer. Both $P_{\mathrm{T}}(n)\mathcal{A}'$ and $n \times \mathcal{A}'$ vanish. The $1/r$ radiation fields and the limiting flux are zero in that direction. The scalar amplitude is needed for this cancellation in the electric field.

Self-check. Why does differentiating an angle-dependent amplitude not add another $1/r$ term to the Cartesian gradient of $\mathcal{V}_f(u, n)/r$?

Answer. Changing x tangentially changes $n = x/r$ at rate $1/r$. The amplitude already has its factor $1/r$, so an angular derivative enters at order r^{-2}. The exact derivative estimate in the proof bounds this term uniformly.

Self-check. For a dipole acceleration $\mathbf{a}$, compute the angular integral of its leading power density $|n \times \mathbf{a}|^2/(16\pi^2)$.

Answer. The integral of $|n \times \mathbf{a}|^2$ is $8\pi|\mathbf{a}|^2/3$. Division by $16\pi^2$ gives $|\mathbf{a}|^2/(6\pi)$. A missing 4π in the retarded potential would change this coefficient.

Self-check. Let f be radial, nonnegative, supported in $|y| \le 1$, with integral one. Put $M_2 = \int |y|^2 f(y) \mathrm{d}^3 y$ and take $\mathbf{p}_0(s) = \mathbf{a} s^4$ for $|s| \le 1$, with a smooth compact cutoff outside this interval and $\mathbf{a} \in \mathbb{R}^3 \setminus \{0\}$. Compute $P(0)$ for $0 < \varepsilon \le 1$, although $\ddot{\mathbf{p}}(0) = 0$.

Answer. Radial symmetry gives $\int (n \cdot y)^2 f(y) \mathrm{d}^3 y = M_2/3$. The sampled acceleration is $12\varepsilon^4 (n \cdot y)^2 \mathbf{a}$, so $\mathcal{A}'(0, n) = \varepsilon^4 M_2 \mathbf{a}/\pi$. The angular integral gives $P(0) = 8\varepsilon^8 M_2^2 |\mathbf{a}|^2/(3\pi) > 0$. A relative error with a zero leading value could not allow this result.